

WEEK1

Name: P.Bindhu Madhavi

Roll Number: 2520030077

Step 1: Create Database

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: My Example* Praticse Example(Week2)* Aggeregate Functions(Week2)* Week1* SQLA

MANAGEMENT

- Server Status
- Client Connectic
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Rep

Administration

Information

No object selected

```
1 2520030077-Bindhu Madhavi
2 CREATE DATABASE IF NOT EXISTS bookflow_db;
3 USE bookflow_db;
4
```

Limit to 1000 rows

Output

Action Output

#	Time	Action	Message
1	19:26:05	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected, 1 warning(s): 1007 Can't c
2	19:26:05	USE bookflow_db	0 row(s) affected
3	19:27:16	CREATE TABLE books (book_id INT AUTO_INCREMENT...	0 row(s) affected

Step 2: Create the books table

MySQL Workbench

The screenshot shows the MySQL Workbench interface with the 'Week1*' tab selected. The SQL editor contains the following code:

```
1 • CREATE DATABASE IF NOT EXISTS bookflow_db;
2 • USE bookflow_db;
3 ✖ 2520030077-Bindhu Madhavi
4 • CREATE TABLE books (
5   book_id INT AUTO_INCREMENT PRIMARY KEY,
6   title VARCHAR(255) NOT NULL,
7   isbn VARCHAR(13) NOT NULL UNIQUE,
8   published_year INT,
9   CONSTRAINT chk_published_year CHECK (published_year < 2027)
10 );
11
```

The left sidebar shows the 'MANAGEMENT' section with 'Server Status' selected. The 'Output' panel at the bottom shows the execution results:

#	Time	Action	Message
1	19:26:05	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected, 1 warning(s): 1007 Can't c
2	19:26:05	USE bookflow_db	0 row(s) affected
3	19:27:16	CREATE TABLE books (book_id INT AUTO_INCREMENT...	0 row(s) affected

Step 3: Create the members table

MySQL Workbench

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File Edit View Query Database Server Tools Scripting Help

Navigator: Query 1 Week2 Week2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Open

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Rep

Administration

Information

No object selected

```

1 • CREATE DATABASE IF NOT EXISTS bookflow_db;
2 • USE bookflow_db;
3 • CREATE TABLE books (
4     book_id INT AUTO_INCREMENT PRIMARY KEY,
5     title VARCHAR(255) NOT NULL,
6     isbn VARCHAR(13) NOT NULL UNIQUE,
7     published_year INT,
8     CONSTRAINT chk_published_year
9     CHECK (published_year < 2027)
10 );
11 ✖ 2520030077-bindhu Madhavi
12 • CREATE TABLE members (
13     member_id INT AUTO_INCREMENT PRIMARY KEY,
14     full_name VARCHAR(100) NOT NULL,
15     email VARCHAR(150) NOT NULL UNIQUE
16 );
17

```

Output

Action Output

#	Time	Action	Message
✓ 5	19:49:03	DROP TABLE IF EXISTS members	0 row(s) affected
✓ 6	19:49:03	DROP TABLE IF EXISTS books	0 row(s) affected
✓ 7	19:49:16	CREATE TABLE books (book_id INT AUTO_INCREM...	0 row(s) affected
✓ 8	19:49:24	CREATE TABLE members (member_id INT AUTO_INC...	0 row(s) affected

Step 4: Inspect the table structures

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Navigator: Query 1 Week2 Week2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Open

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Report

Administration

Information: No object selected

```
2520030077-bindhu Madhavi
CREATE TABLE members (
  member_id INT AUTO_INCREMENT PRIMARY KEY,
  full_name VARCHAR(100) NOT NULL,
  email VARCHAR(150) NOT NULL UNIQUE
);
DESCRIBE books;
```

Result Grid

	Field	Type	Null	Key	Default	Extra
▶	book_id	int	NO	PRI	NULL	auto_increment
	title	varchar(255)	NO		NULL	
	isbn	varchar(13)	NO	UNI	NULL	
	published_year	int	YES		NULL	

Result 1 x

Output

Action Output

#	Time	Action	Message
✓ 6	19:49:03	DROP TABLE IF EXISTS books	0 row(s) affected
✓ 7	19:49:16	CREATE TABLE books (book_id INT AUTO_INCREM...	0 row(s) affected
✓ 8	19:49:24	CREATE TABLE members (member_id INT AUTO_IN...	0 row(s) affected
✓ 9	19:50:35	DESCRIBE books	4 row(s) returned

MySQL Workbench

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Navigator

MANAGEMENT

- Server Status
- Client Connection
- Users and Privileges
- Status and System
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Report

Administration

Information

No object selected

Query 1 Week2 Week2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Op

```

12 CREATE TABLE members (
13     member_id INT AUTO_INCREMENT PRIMARY KEY,
14     full_name VARCHAR(100) NOT NULL,
15     email VARCHAR(150) NOT NULL UNIQUE
16 );
17 DESCRIBE books;
18 DESCRIBE members;

```

Limit to 1000 rows

Result Grid

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI	NULL	auto_increment
full_name	varchar(100)	NO		NULL	
email	varchar(150)	NO	UNI	NULL	

Result 2 x

Output

Action Output

#	Time	Action	Message
7	19:49:16	CREATE TABLE books (book_id INT AUTO_INCREM...	0 row(s) affected
8	19:49:24	CREATE TABLE members (member_id INT AUTO_IN...	0 row(s) affected
9	19:50:35	DESCRIBE books	4 row(s) returned
10	19:51:29	DESCRIBE members	3 row(s) returned

Step 5: Insert 3 books

MySQL Workbench

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Navigator: Query 1 Week2 Week2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Open

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Report

Administration

Information: No object selected

```

4      book_id INT AUTO_INCREMENT PRIMARY KEY,
5      title VARCHAR(255) NOT NULL,
6      isbn VARCHAR(13) NOT NULL UNIQUE,
7      published_year INT,
8      CONSTRAINT chk_published_year
9      CHECK (published_year < 2027)
10 );
11
12 CREATE TABLE members (
13     member_id INT AUTO_INCREMENT PRIMARY KEY,
14     full_name VARCHAR(100) NOT NULL,
15     email VARCHAR(150) NOT NULL UNIQUE
16 );
17 DESCRIBE books;
18 DESCRIBE members;
19 2520030077-bindhu Madhavi
20 INSERT INTO books (title, isbn, published_year)
21 VALUES
22 ('The Alchemist','9780061122415',1988),
23 ('Clean Code','9780132350884',2008),
24 ('Atomic Habits','9780735211292',2018);

```

Output

Action Output

#	Time	Action	Message
8	19:49:24	CREATE TABLE members (member_id INT AUTO_IN...	0 row(s) affected
9	19:50:35	DESCRIBE books	4 row(s) returned
10	19:51:29	DESCRIBE members	3 row(s) returned
11	19:52:17	INSERT INTO books (title, isbn, published_year) VALUES...	3 row(s) affected Records: 3 Duplicate

Step 6: Insert 3 members

MySQL Workbench

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File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Rep

Administration

Information

No object selected

Query 1 Week2 Week2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Oper

Limit to 1000 rows

```

11
12 CREATE TABLE members (
13     member_id INT AUTO_INCREMENT PRIMARY KEY,
14     full_name VARCHAR(100) NOT NULL,
15     email VARCHAR(150) NOT NULL UNIQUE
16 );
17 DESCRIBE books;
18 DESCRIBE members;
19 2520030077-bindhu Madhavi
20 INSERT INTO books (title, isbn, published_year)
21 VALUES
22 ('The Alchemist', '9780061122415', 1988),
23 ('Clean Code', '9780132350884', 2008),
24 ('Atomic Habits', '9780735211292', 2018);
25
26 SELECT * FROM books;
27 INSERT INTO members (full_name, email)
28 VALUES
29 ('Bindhu Madhavi', 'bindhu.madhavi@example.com'),
30 ('Priya Sharma', 'priya.sharma@example.com'),
31 ('Ravi Verma', 'ravi.verma@example.com');

```

Output

Action Output

#	Time	Action	Message
9	19:50:35	DESCRIBE books	4 row(s) returned
10	19:51:29	DESCRIBE members	3 row(s) returned
11	19:52:17	INSERT INTO books (title, isbn, published_year) VALUES...	3 row(s) affected Records: 3 Duplicates: 0
12	19:53:35	INSERT INTO members (full_name, email) VALUES ('Bind...	3 row(s) affected Records: 3 Duplicates: 0

Object Info

Step 7:

MySQL Workbench

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File Edit View Query Database Server Tools Scripting Help

MANAGEMENT

Server Status

Client Connectio

Users and Privile

Status and Syste

Data Export

Data Import/Res

INSTANCE

Startup / Shutdc

Server Logs

Options File

PERFORMANCE

Dashboard

Performance Rep

Administration

Information

No object selected

Query 1 Week2 Week2 (DDL)* DCL Commands* Weeks2(TCL)* Joint

26 • SELECT * FROM books;

27 • INSERT INTO members (full_name, email)

28 VALUES

29 ('Bindhu Madhavi','bindhu.madhavi@example.com'),

30 ('Priya Sharma','priya.sharma@example.com'),

31 ('Ravi Verma','ravi.verma@example.com');

32 • SELECT * FROM books;

Result Grid

	book_id	title	isbn	published_year
▶	1	The Alchemist	9780061122415	1988
	2	Clean Code	9780132350884	2008
	3	Atomic Habits	9780735211292	2018
*	NULL	NULL	NULL	NULL

books 3 x

Output

Action Output

#	Time	Action	Message
✓ 10	19:51:29	DESCRIBE members	3 row(s) returned
✓ 11	19:52:17	INSERT INTO books (title, isbn, published_year) VALUES...	3 row(s) affected Rec
✓ 12	19:53:35	INSERT INTO members (full_name, email) VALUES ('Bind...	3 row(s) affected Rec
✓ 13	19:54:27	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: Query 1 Week2 Week2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Open

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Report

Administration

Information

No object selected

```
27 • INSERT INTO members (full_name, email)
28 VALUES
29 ('Bindhu Madhavi','bindhu.madhavi@example.com'),
30 ('Priya Sharma','priya.sharma@example.com'),
31 ('Ravi Verma','ravi.verma@example.com');
32 • SELECT * FROM books;
33 • SELECT * FROM members;
```

Result Grid

	member_id	full_name	email
▶	1	Bindhu Madhavi	bindhu.madhavi@example.com
	2	Priya Sharma	priya.sharma@example.com
	3	Ravi Verma	ravi.verma@example.com
*	NULL	NULL	NULL

members 4 x

Output

Action Output

#	Time	Action	Message
✓ 11	19:52:17	INSERT INTO books (title, isbn, published_year) VALUES...	3 row(s) affected Records: 3 Duplicates: 0
✓ 12	19:53:35	INSERT INTO members (full_name, email) VALUES ('Bind...	3 row(s) affected Records: 3 Duplicates: 0
✓ 13	19:54:27	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned
✓ 14	19:54:59	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned

Step 8: Constraint Tests

MySQL Workbench

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File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdo
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Rep

Administration

Information

No object selected

Query 1 Week2 Week2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Open

Limit to 1000 rows

```

17 • DESCRIBE books;
18 • DESCRIBE members;
19 ✖ 2520030077-bindhu Madhavi
20 INSERT INTO books (title, isbn, published_year)
21 VALUES
22 ('The Alchemist','9780061122415',1988),
23 ('Clean Code','9780132350884',2008),
24 ('Atomic Habits','9780735211292',2018);
25
26 • SELECT * FROM books;
27 • INSERT INTO members (full_name, email)
28 VALUES
29 ('Bindhu Madhavi','bindhu.madhavi@example.com'),
30 ('Priya Sharma','priya.sharma@example.com'),
31 ('Ravi Verma','ravi.verma@example.com');
32 • SELECT * FROM books;
33 • SELECT * FROM members;
34 • INSERT INTO books (title, isbn, published_year)
35 VALUES
36 ('Fake Copy','9780061122415',2000);
37

```

Output

Action Output

#	Time	Action	Message
✓ 12	19:53:35	INSERT INTO members (full_name, email) VALUES ('Bind...	3 row(s) affected Records: 3 Duplicat
✓ 13	19:54:27	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned
✓ 14	19:54:59	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned
✖ 15	19:57:00	INSERT INTO books (title, isbn, published_year) VALUES...	Error Code: 1062. Duplicate entry '978

Test 1: Duplicate ISBN

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Report

Administration

Information

No object selected

Query 1

```

19 2520030077--bindhu Madhavi
20 INSERT INTO books (title, isbn, published_year)
21 VALUES
22 ('The Alchemist','9780061122415',1988),
23 ('Clean Code','9780132350884',2008),
24 ('Atomic Habits','9780735211292',2018);
25
26 SELECT * FROM books;
27 INSERT INTO members (full_name, email)
28 VALUES
29 ('Bindhu Madhavi','bindhu.madhavi@example.com'),
30 ('Priya Sharma','priya.sharma@example.com'),
31 ('Ravi Verma','ravi.verma@example.com');
32 SELECT * FROM books;
33 SELECT * FROM members;
34 INSERT INTO books (title, isbn, published_year)
35 VALUES
36 ('Fake Copy','9780061122415',2000);
37 INSERT INTO books (title, isbn, published_year)
38 VALUES
39 (NULL, '9999999999999',2010);
40

```

Output

Action Output

#	Time	Action	Message
13	19:54:27	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned
14	19:54:59	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned
15	19:57:00	INSERT INTO books (title, isbn, published_year) VALUES...	Error Code: 1062. Duplicate
16	19:58:01	INSERT INTO books (title, isbn, published_year) VALUES...	Error Code: 1048. Column t

Test 2: NULL title

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: Query 1 Week2 Week2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Open SQLAddit

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdo
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Rep

Administration

Information

No object selected

```

23 ('Clean Code','9780132350884',2008),
24 ('Atomic Habits','9780735211292',2018);
25
26 • SELECT * FROM books;
27 • INSERT INTO members (full_name, email)
28 VALUES
29 ('Bindhu Madhavi','bindhu.madhavi@example.com'),
30 ('Priya Sharma','priya.sharma@example.com'),
31 ('Ravi Verma','ravi.verma@example.com');
32 • SELECT * FROM books;
33 • SELECT * FROM members;
34 • INSERT INTO books (title, isbn, published_year)
35 VALUES
36 ('Fake Copy','9780061122415',2000);
37 • INSERT INTO books (title, isbn, published_year)
38 VALUES
39 (NULL,'999999999999',2010);
40 • INSERT INTO books (title, isbn, published_year)
41 VALUES
42 ('Time Traveler','888888888888',2030);
43

```

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Output

Action Output

#	Time	Action	Message
✓ 14	19:54:59	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned
✗ 15	19:57:00	INSERT INTO books (title, isbn, published_year) VALUES...	Error Code: 1062. Duplicate entry '9780061122415'
✗ 16	19:58:01	INSERT INTO books (title, isbn, published_year) VALUES...	Error Code: 1048. Column 'title' cannot be null
✗ 17	19:58:40	INSERT INTO books (title, isbn, published_year) VALUES...	Error Code: 3819. Check constraint 'chk_published_'

Object Info

Test 3: Future publication year (CHECK violation) &&
Duplicate email (UNIQUE violation)

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: Query 1 Week2 Week2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Oper SQLA

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdo
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Rep

Administration Administration

Information No object selected

```
22 ('The Alchemist','9780061122415',1988),
23 ('Clean Code','9780132350884',2008),
24 ('Atomic Habits','9780735211292',2018);
25
26 • SELECT * FROM books;
27 • INSERT INTO members (full_name, email)
28 VALUES
29 ('Bindhu Madhavi','bindhu.madhavi@example.com'),
30 ('Priya Sharma','priya.sharma@example.com'),
31 ('Ravi Verma','ravi.verma@example.com');
32 • SELECT * FROM books;
33 • SELECT * FROM members;
34 • INSERT INTO books (title, isbn, published_year)
35 VALUES
36 ('Fake Copy','9780061122415',2000);
37 • INSERT INTO books (title, isbn, published_year)
38 VALUES
39 (NULL,'9999999999999',2010);
40 • INSERT INTO members (full_name,email)
41 VALUES
42 ('Anil Clone','anil.kumar@example.com');
```

Output

Action Output

#	Time	Action	Message
✖ 15	19:57:00	INSERT INTO books (title, isbn, published_year) VALUES...	Error Code: 1062. Duplicate entry '9780061122
✖ 16	19:58:01	INSERT INTO books (title, isbn, published_year) VALUES...	Error Code: 1048. Column 'title' cannot be null
✖ 17	19:58:40	INSERT INTO books (title, isbn, published_year) VALUES...	Error Code: 3819. Check constraint 'chk_public
✔ 18	19:59:15	INSERT INTO members (full_name,email) VALUES ('Anil ...	1 row(s) affected